Exercises: Multivariate Taylor Expansion, Critical Points & Extrema

Mathematical Foundations for Machine Learning

Taylor Expansion — Computation

Exercise 1.1 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as $f(x, y) = e^{x+y}$. Compute the second-order Taylor expansion of f around $x_0 = (0, 0)$.

Exercise 1.2 [Easy]
Let $f(x, y) = \sin(x)\cos(y)$. Compute the second-order Taylor expansion around $(0, 0)$.

Exercise 1.3 [Medium]
Given $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(x) = \sqrt{1 + \|x\|^2}$. Compute the second-order Taylor expansion at $x_0 = 0$.

(Exam 2026, Exercise 2b)

Exercise 1.4 [Medium]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(x, y) = \log(1+x^2+y^2+xy)$. Compute the second-order Taylor expansion around $(0, 0)$ and verify your result by comparing with the expansion of $\log(1 + u)$ for small u.

Exercise 1.5 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive definite. Define $f : \mathbb{R}^n \to \mathbb{R}$ by $f(x) = \log(x^T Ax)$. Compute the second-order Taylor expansion around any point x_0 with $x_0^T Ax_0 \neq 0$.

Critical Points — Finding and Classifying

Exercise 2.1 [Easy]
Find all critical points of $f(x, y) = x^2 + y^2 - 2x - 4y + 5$ and classify them as local min, local max, or saddle point.

Exercise 2.2 [Easy]
Find and classify all critical points of $f(x, y) = x^3 - 3xy + y^3$.

Exercise 2.3 [Medium]
Find and classify all critical points of $f(x, y) = (x^2 + y^2)e^{-(x^2+y^2)}$.

-> lots of computation, only to practice algebra, the one below better for recall

Exercise 2.4 [Medium]
Consider the function from Exam 2021 Exercise 2a in a related form: let $f(x, y) = x^2 + xy + y^2 - 3x - 3y$. Find all critical points, classify them, and compute the value of f at each.

Then compute the directional derivative at the critical point in the direction $v = \begin{pmatrix}1\\2\end{pmatrix}$ and explain why the result makes sense.

no first order change != no change

Exercise 2.5 [Hard]
Consider $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = x^2 + y^2(1 - x)^3$.

(a) Find all critical points and classify them using the Hessian.

(b) Show that f has no global minimum by finding a direction along which $f \to -\infty$.

(c) Does this contradict the result that, for $f \in C^1(\mathbb{R})$, a unique critical point that is a local minimum must be the global minimum? Explain carefully.

Local vs. Global Extrema

When hessian eigenvalues are 0 that means that the second grade curvature is 0. But it could be that the higher order curvature be neg or pos. (when we have second order curvature it shadows the higher order curvature, because since we do taylor, we are talking about a tiny change, 0.00x, so very tiny^2 > tiny^3, 4...

Exercise 3.1 [Easy]
State the necessary condition and the sufficient condition for a critical point x_0 of $f \in C^2(\Omega)$ to be a local minimum. Give a concrete example where the necessary condition is satisfied but the sufficient condition is not.

Exercise 3.2 [Medium]
Let $f \in C^1(\mathbb{R})$ have a single critical point x_0 which is a local minimum. Prove that x_0 is the global minimum.

Hint: Use the theorem of Rolle by contradiction.
(Exam 2021 style — proof question)

Exercise 3.3 [Medium]
Prove that any local minimum of a convex function $f : \Omega \to \mathbb{R}$ over a convex set Ω is also a global minimum.

Hint: Prove by contradiction using the definition of convexity.
(Exam 2021, Exercise 3a)

Exercise 3.4 [Hard]
Let $f \in C^1(\mathbb{R}^n)$ have a unique critical point x_0 which is a local minimum. Suppose additionally that there exists $r > 0$ such that $f(x) > f(x_0)$ for all $x \in \mathbb{R}^n \setminus B(x_0, r)$. Prove that x_0 is the global minimum.

Hint: Use compactness of the closed ball and the extreme value theorem.

Positive Semi-Definite Hessians and Degenerate Cases

Exercise 4.1 [Easy]
Classify the critical point at the origin for each of the following functions:

(a) $f(x, y) = x^2 + y^4$

(b) $f(x, y) = x^2 - y^4$

(c) $f(x, y) = x^4 + y^4$

Note that the Hessian is only semi-definite at the origin for each. Can the Hessian test conclude anything? Determine the nature of each critical point from the definition.

Exercise 4.2 [Medium]
Let $\|u\| = 1$ and define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \langle x, (I - uu^T)x \rangle - \langle u, x \rangle^4.$$

Show that $\nabla f(0) = 0$ and $Hf(0) \succeq 0$, but $x_0 = 0$ is **not** a local minimum.
(This is the counterexample from the lecture notes.)

To prove one direction then take alfa * u

Exercise 4.3 [Hard]
Let $f \in C^2(\mathbb{R}^n)$ with $\nabla f(x_0) = 0$. Prove the following:

(a) (Necessary condition) If x_0 is a local minimum, then $Hf(x_0) \succeq 0$.

(b) (Sufficient condition) If $Hf(x_0) \succ 0$, then x_0 is a strict local minimum.

Use the second-order Taylor expansion with remainder.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Let $W \in \mathbb{R}^{m\times n}$ and define the loss function of a single-layer linear network:

$$L(W) = \|y - Wx\|_2^2$$

where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ are fixed.

(a) Compute $\nabla_W L$ and find all critical points.

(b) Compute the Hessian. *Hint:* treat W as a vector in $\mathbb{R}^{mn}$ via $\mathrm{vec}(W)$, and use the identity $\mathrm{vec}(ABC) = (C^T \otimes A)\mathrm{vec}(B)$.

(c) Classify the critical point(s). Under what condition on x is the critical point a strict local minimum?

Exercise 5.2 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Consider the function $f : \mathbb{R}^n \to \mathbb{R}$:

$$f(x) = \frac{1}{2}x^T Ax - b^T x$$

where $b \in \mathbb{R}^n$ is fixed.

(a) Find the unique critical point and prove it is a strict global minimum.

(b) Compute the second-order Taylor expansion of f around the minimizer x^*.

(c) Prove that $f(x) - f(x^*) = \frac{1}{2}(x - x^*)^T A(x - x^*)$ for all x. Interpret this geometrically in terms of the level sets of f.

Exercise 5.3 [Hard]
Let $f \in C^2(\mathbb{R}^n)$ and suppose x_0 is a critical point with $Hf(x_0)$ having eigenvalues $\lambda_1 < 0 < \lambda_2$, with corresponding eigenvectors u_1, u_2. Using the Taylor expansion:

(a) Prove that x_0 is a saddle point by showing that f increases along one direction and decreases along another.

(b) In the context of training a neural network with loss function $L(\theta)$, explain why a saddle point might be more problematic than a local minimum for gradient descent. How does the second-order information from Hf help?